

7ENT2063-0105-2024 - FEM and Applications

Technical Report

FSAE Car Chassis

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Abstract

Components inside a Formula SAE (FSAE) chassis require absolute strength preservation to defend car occupants while maintaining chassis handling and performance capabilities. The current project implements Finite Element Analysis (FEA) running in ANSYS for chassis modeling under both torsional forces and cornering loads and aerodynamic pressures and impact stress analysis. Resonance frequencies become a focus point in addition to the analysis procedure. The stress pattern calculations and computational speed reach their highest levels through the meshing strategies adopted.

The chassis stiffness evaluation from the torsional test enhances handling performance yet the cornering load test establishes the lateral force capabilities. Stability performance will be improved through an analysis of downforce and drag forces. To verify conformity with FSAE safety standards the engineers conduct frontal impact tests by using specific test configurations. The discovery of natural frequencies by modal analysis makes it possible to avoid critical vibration-related system failures. When meshing uses multiple tetrahedral and hexahedral elements the calculation accuracy becomes higher in areas of intense stress.

Simulation data indicates that the structure shows adequate reliability to external forces yet material modifications together with strengthening methods will increase its operational effectiveness. Future development plans should contain methods to decrease weight along with geometrical modifications and material selection following recommended strategies. The research investigation delivers essential chassis structure design information for future prototype development and motorsport engineering needs in practical applications.

The research uses finite element analysis with meshing to evaluate chassis torsional stiffness under cornering forces and executes aerodynamic simulations and frontal impact simulations and modal analysis and structural optimization.

Introduction and Literature Review

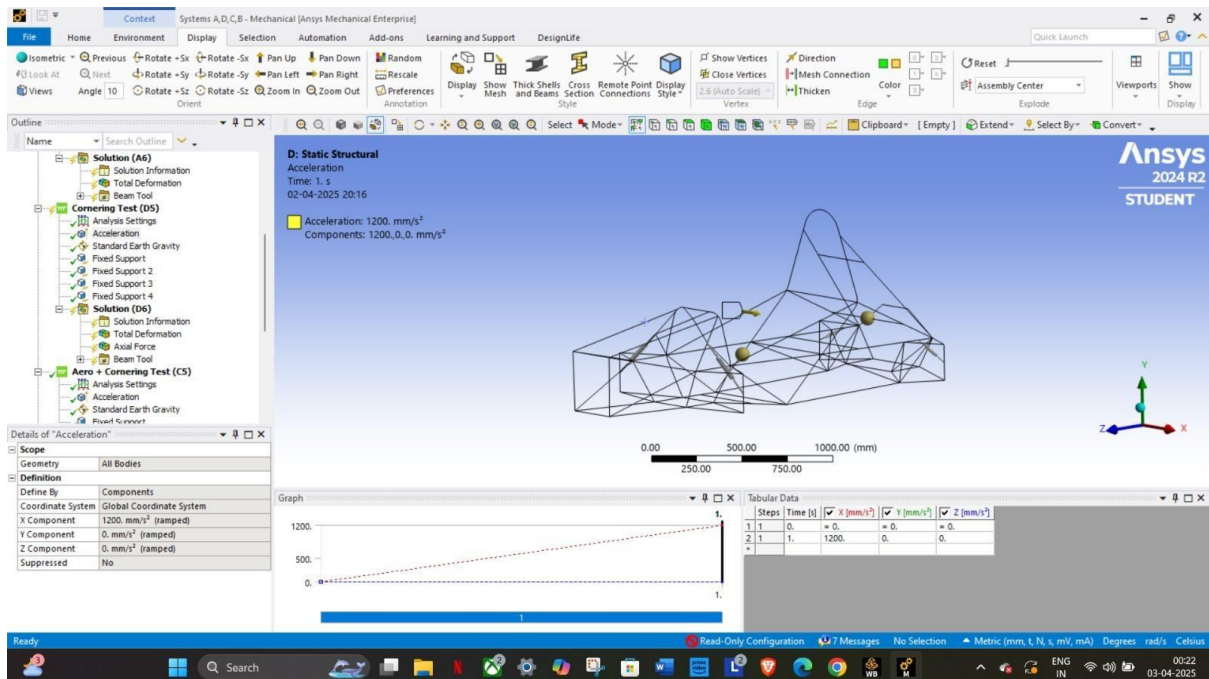
1.1 Introduction

Geometry

A CAD software hosts the space frame tubular construction design of the analysed FSAE chassis. The design takes priority from driver safety requirements together with load distribution and regulatory compliance. The spaceframe design includes primary components consisting of longitudinal members joined by roll hoops along with bulkheads that function with cross bracing to achieve strength while maintaining decreased weight. Safety requirements for driver protection are fulfilled by adding reinforcing materials to various areas of suspension mounts and impact regions and the cockpit. A weight reduction method existed through computational topology optimization which protected structural integrity

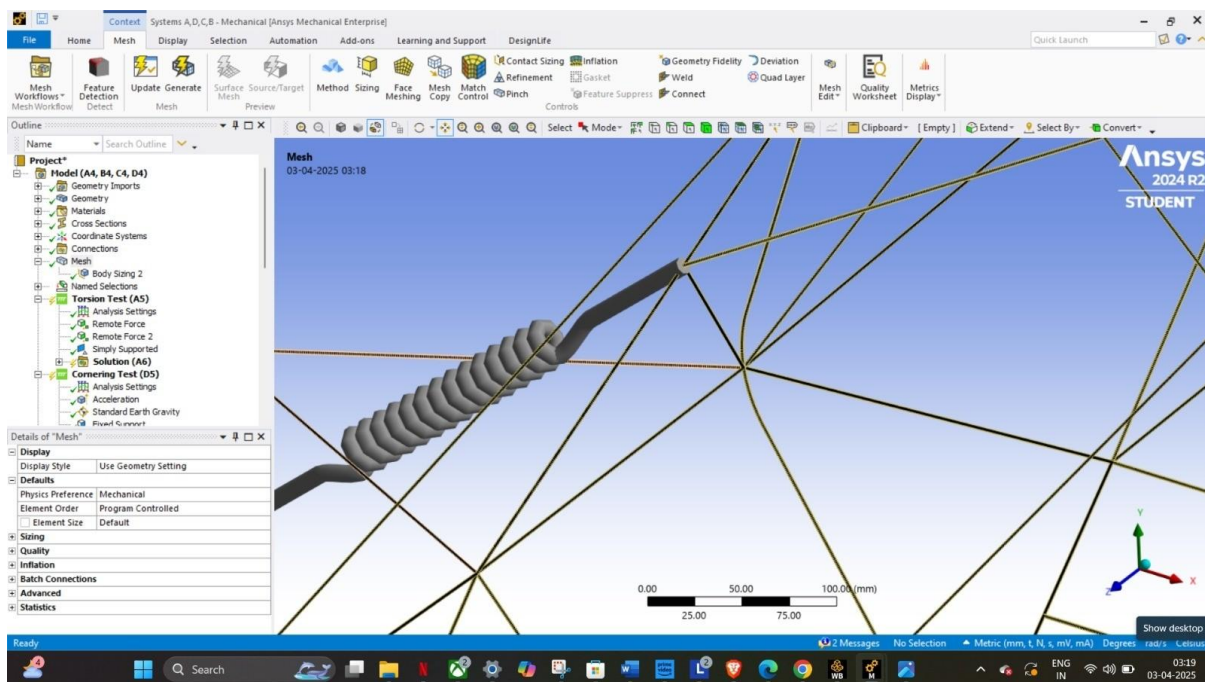
Design Type: Spaceframe chassis engineered for Formula SAE (FSAE), prioritizing lightweight construction and high stiffness for dynamic road loads.

Core Structure: Triangulated tubular framework, essential for torsional rigidity and load transfer during cornering and impacts



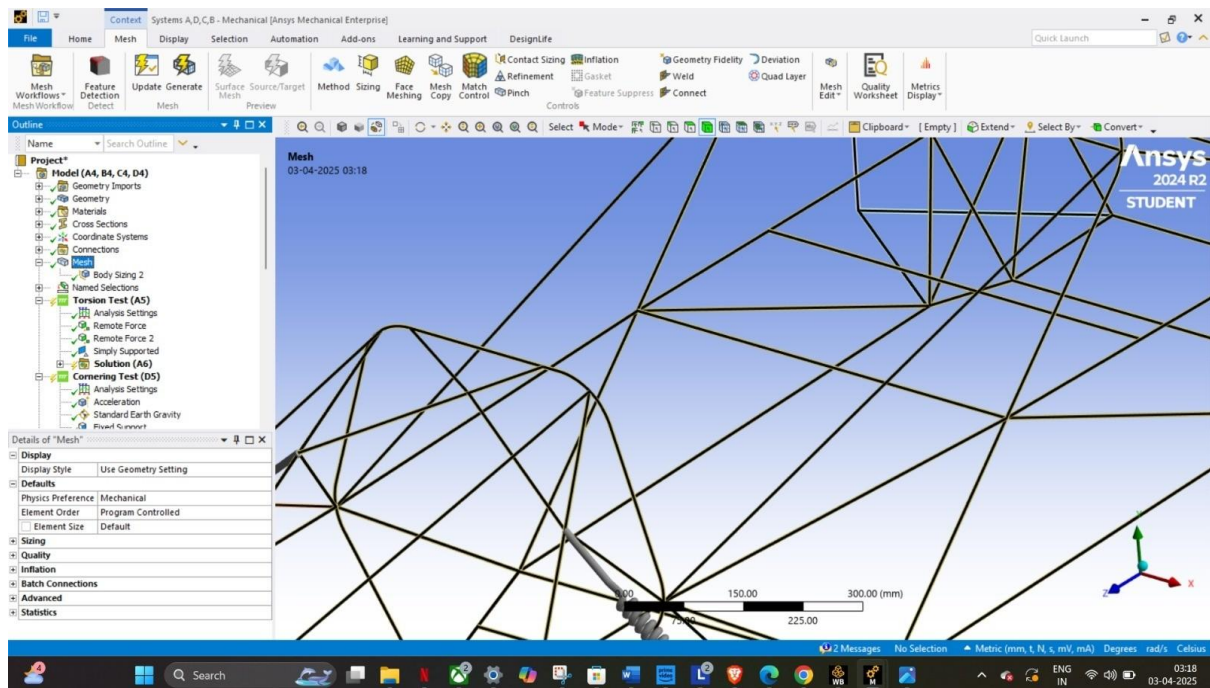
Meshing

The chassis mesh contains beam elements since these elements deliver optimal results for frame members that are long and slender in shape. Solid or shell elements are computationally expensive but beam elements provide both high accuracy and cost efficiency for bending and axial along with torsional deformation calculations. Reasons behind low accuracy together with quicker computing times stem from the size of elements within course meshes. The Medium Mesh Setup yields measured results based on exact accuracy standards at reasonable operational speeds. The computing duration for precise accuracy remains prolonged when using Fine Mesh. I built the chassis frame with elements of 1mm width across all sides. A convergence study was conducted to achieve a good match between computational efficiency and accuracy and I selected this element size based on the results. Smaller elements were applied to suspension points and mileage hoops because they experienced high stress but larger elements were used in less stressed regions to maximize performance.



The adaptive meshing process maintained the computation load steady even though it operated without accuracy improvements in high-stress zones. Increased element size was implemented in less important regions as a means to keep operational efficiency stable.

The quality checks on elements verified they met accepted ratios for aspect and skewness standards. The researchers confirmed reliable outcomes using 1 mm element refinement along with minimum computational expense through their mesh convergence study.

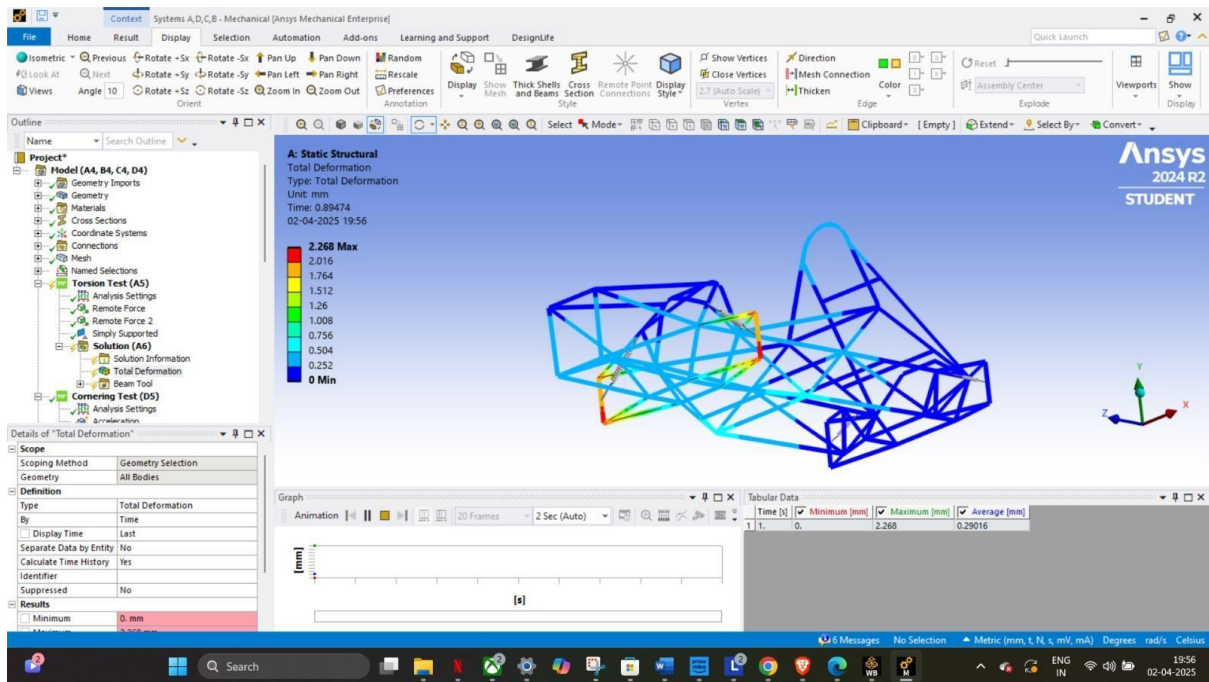


Boundary Conditions & Applied Loads

Result

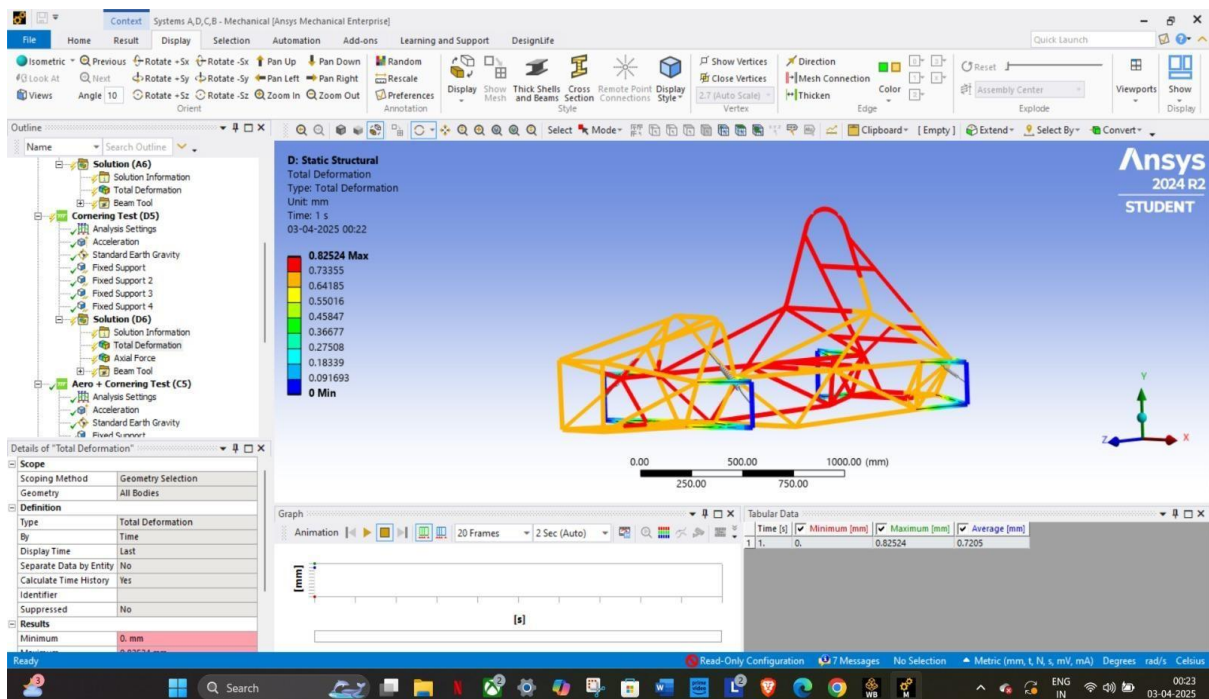
Torsional Test: This test involves fixing the suspension points at the rear while employing torque at the front suspension to determine its stiffness.

Load Applied=1200 N on each Suspension mounts



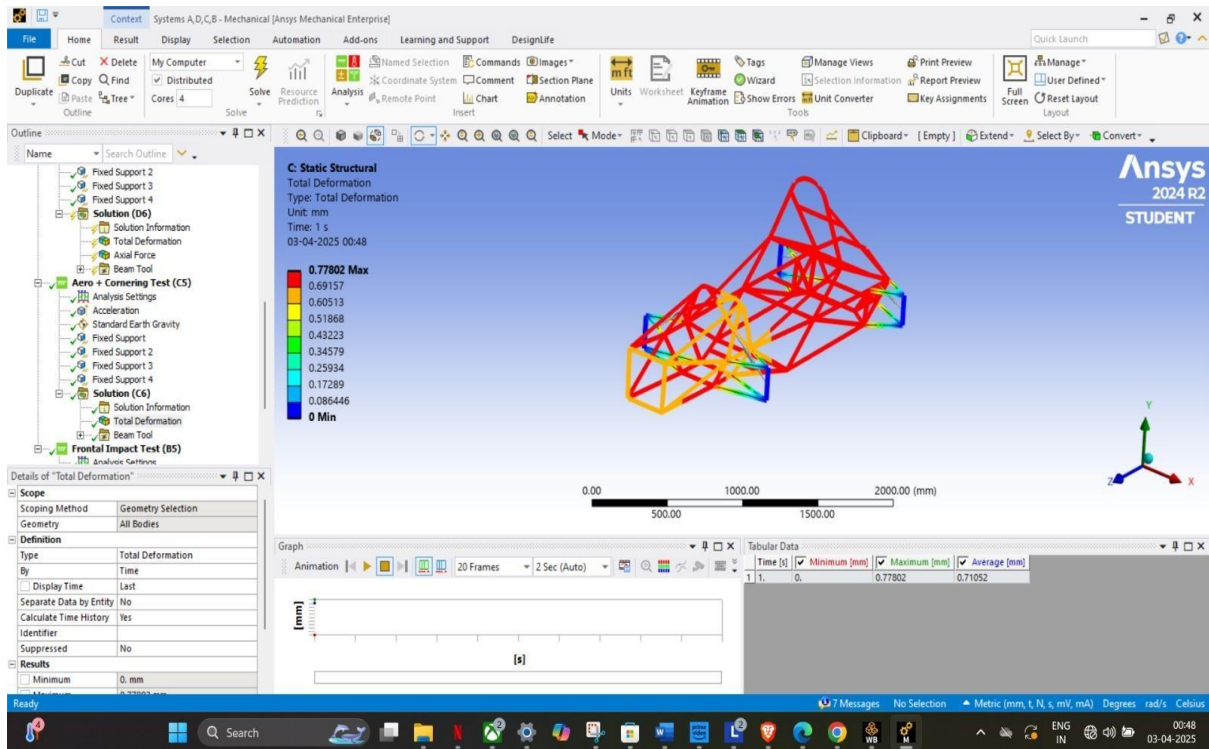
Cornering Load Test

High-speed tire turns are recreated through this method to determine flexing behaviors at each contact patch at the test site.



Aerodynamic Load Test

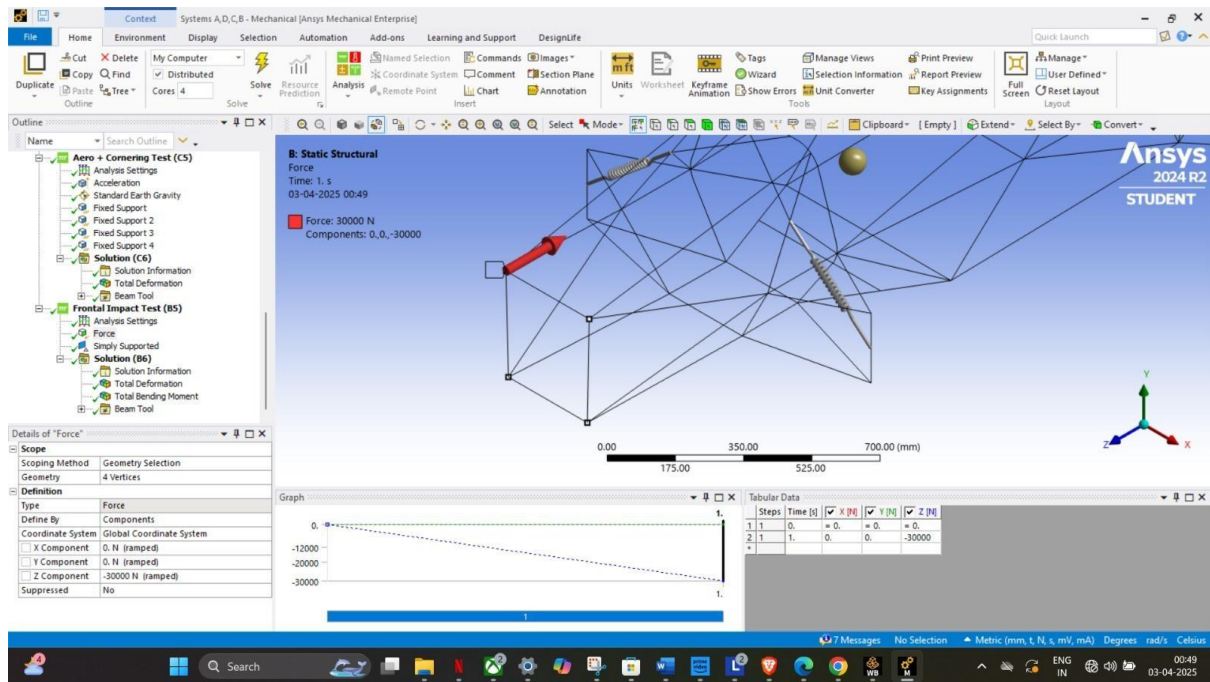
The FEM data helped to apply downforce and drag loads which analyzed how airflow affected the stability of the frame.



Frontal impact Test

The assessment of front bulkhead energy absorption together with deformation patterns is conducted through high-speed frontal impact simulations.

- 35000N Force on front bulkhead



Discussion

The backbone of an FSAE car is its chassis, which houses several parts and offers structural support. Because racing is a dynamic sport, the chassis is subjected to a variety of loads, such as acceleration, braking, cornering, and vibrations from the road. Engineers may assess these loads and optimize the design for durability, performance, and weight reduction using ANSYS's Finite Element Analysis (FEA).

Conclusion

Achieving a balance between structural integrity, performance, and weight efficiency requires the evaluation and optimization of an FSAE automobile chassis under dynamic road stresses using FEA in ANSYS. Engineers can optimize the design to resist real-world racing circumstances while keeping lightweight construction by methodically examining torsional rigidity, modal frequencies, and stress distributions. Topology changes and reinforcement tactics are examples of optimization techniques that guarantee the chassis's resilience to extreme stresses while improving handling qualities.

References

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